

Volume & Surface Area — Practice Questions

Digital SAT Style | Mixed Difficulty | Total Questions: 25

Q1.

DIFFICULTY: EASY

A rectangular prism has dimensions 4 cm $\times$ 5 cm $\times$ 6 cm. What is its volume?

- A) 15 cm³
 - B) 60 cm³
 - C) 120 cm³
 - D) 240 cm³
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Q2.

DIFFICULTY: EASY

A cube has a surface area of 150 cm². What is the length of each edge?

- A) 5 cm
 - B) 6 cm
 - C) 10 cm
 - D) 25 cm
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Q3.

DIFFICULTY: EASY

A cylinder has radius 3 cm and height 10 cm. What is its volume? (Use $\pi \approx 3.14$)

- A) 28.26 cm³
 - B) 94.2 cm³
 - C) 282.6 cm³
 - D) 942 cm³
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Q4.

DIFFICULTY: EASY

A square pyramid has a base edge of 6 cm and a height of 10 cm. What is its volume?

- A) 60 cm³
 - B) 120 cm³
 - C) 180 cm³
 - D) 360 cm³
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Q5.

DIFFICULTY: EASY

A sphere has radius 3 cm. What is its volume? (Use $\pi \approx 3.14$)

- A) 28.26 cm³

- B) 56.52 cm^3
 - C) 113.04 cm^3
 - D) 904.32 cm^3
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Q6.

DIFFICULTY: MEDIUM

A cone has radius 6 cm and slant height 10 cm. What is its height?

- A) 4 cm
 - B) 8 cm
 - C) 12 cm
 - D) 16 cm
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Q7.

DIFFICULTY: MEDIUM

A cylinder has volume $500\pi \text{ cm}^3$ and height 20 cm. What is its radius?

- A) 5 cm
 - B) 10 cm
 - C) 15 cm
 - D) 25 cm
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Q8.

DIFFICULTY: MEDIUM

If all dimensions of a rectangular prism are tripled, by what factor does the volume increase?

- A) 3
 - B) 9
 - C) 27
 - D) 81
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Q9.

DIFFICULTY: MEDIUM

A cone has radius 4 cm and height 9 cm. What is its volume? (Use $\pi \approx 3.14$)

- A) $48\pi \text{ cm}^3$
 - B) 150.72 cm^3
 - C) 452.16 cm^3
 - D) 1356.48 cm^3
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Q10.

DIFFICULTY: MEDIUM

A rectangular prism has dimensions $3 \text{ cm} \times 4 \text{ cm} \times 5 \text{ cm}$. What is its surface area?

- A) 47 cm^2

- B) 60 cm^2
 - C) 94 cm^2
 - D) 120 cm^2
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Q11.

DIFFICULTY: MEDIUM

A hemisphere has radius 6 cm. What is its total surface area (including the flat circular base)?

- A) $36\pi \text{ cm}^2$
 - B) $72\pi \text{ cm}^2$
 - C) $108\pi \text{ cm}^2$
 - D) $144\pi \text{ cm}^2$
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Q12.

DIFFICULTY: MEDIUM

Two similar cylinders have radii in ratio 2:3. What is the ratio of their volumes?

- A) 2:3
 - B) 4:9
 - C) 8:27
 - D) 16:81
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Q13.

DIFFICULTY: MEDIUM

A cylinder has radius 5 cm and height 12 cm. What is its lateral (curved) surface area?

- A) $60\pi \text{ cm}^2$
 - B) $120\pi \text{ cm}^2$
 - C) $170\pi \text{ cm}^2$
 - D) $240\pi \text{ cm}^2$
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Q14.

DIFFICULTY: HARD

A cone has radius 3 cm and height 4 cm. What is the slant height?

- A) 4 cm
 - B) 5 cm
 - C) 6 cm
 - D) 7 cm
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Q15.

DIFFICULTY: HARD

A sphere is inscribed in a cube of side 10 cm. What is the volume of the sphere? (Use $\pi \approx 3.14$)

- A) $166.7\pi \text{ cm}^3$

- B) $500/3 \pi \text{ cm}^3$
 - C) $250\pi \text{ cm}^3$
 - D) $1000\pi \text{ cm}^3$
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Q16.

DIFFICULTY: HARD

A cylinder's height is 3 times its radius. If the volume is $192\pi \text{ cm}^3$, what is the radius?

- A) 3 cm
 - B) 4 cm
 - C) 6 cm
 - D) 8 cm
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Q17.

DIFFICULTY: HARD

A cone has the same base and height as a cylinder with volume 300 cm^3 . What is the cone's volume?

- A) 100 cm^3
 - B) 150 cm^3
 - C) 200 cm^3
 - D) 900 cm^3
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Q18.

DIFFICULTY: HARD

A rectangular tank is 8 m long, 6 m wide, and 5 m deep. Water fills it to a depth of 3 m. What volume of water is needed to fill it completely?

- A) 48 m^3
 - B) 96 m^3
 - C) 144 m^3
 - D) 240 m^3
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Q19.

DIFFICULTY: HARD

A solid sphere of radius 9 cm is melted and recast into smaller spheres of radius 3 cm. How many small spheres are formed?

- A) 3
 - B) 9
 - C) 18
 - D) 27
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Q20.

DIFFICULTY: HARD

A cylinder has radius 7 cm and height 10 cm. What is its total surface area? (Use $\pi = 22/7$)

- A) 440 cm^2
 - B) 528 cm^2
 - C) 616 cm^2
 - D) 748 cm^2
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Q21.

DIFFICULTY: HARD

A frustum is formed by cutting a cone of height 12 cm and radius 6 cm at half its height. What is the volume of the frustum? (Use $\pi \approx 3.14$)

- A) $42\pi \text{ cm}^3$
 - B) $63\pi \text{ cm}^3$
 - C) $84\pi \text{ cm}^3$
 - D) $126\pi \text{ cm}^3$
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Q22.

DIFFICULTY: HARD

A cube has edge length 8 cm. A sphere is circumscribed around it (smallest sphere containing the cube). What is the radius of the sphere?

- A) 4 cm
 - B) $4\sqrt{2}$ cm
 - C) $4\sqrt{3}$ cm
 - D) 8 cm
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Q23.

DIFFICULTY: HARD

A triangular prism has a triangular cross-section with base 6 cm and height 4 cm. The prism is 15 cm long. What is its volume?

- A) 90 cm^3
 - B) 120 cm^3
 - C) 180 cm^3
 - D) 360 cm^3
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Q24.

DIFFICULTY: HARD

A hollow pipe is 20 cm long with outer radius 5 cm and inner radius 3 cm. What is the volume of the material in the pipe?

- A) $160\pi \text{ cm}^3$
- B) $320\pi \text{ cm}^3$
- C) $480\pi \text{ cm}^3$

D) $640\pi \text{ cm}^3$

Q25.

DIFFICULTY: HARD

A solid metal cube of edge 12 cm is melted and recast into a cylinder of radius 6 cm. What is the height of the cylinder? (Use $\pi \approx 3.14$)

- A) 12 cm
 - B) 15.28 cm
 - C) 24 cm
 - D) 48 cm
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